

Practical - 04

1. Identify Data Sources

- Network data: Call Detail Records (CDRs), traffic logs, signal strength, latency.
- Customer data: CRM systems, billing, subscriptions, churn records.
- Operational data: Fault management, service quality, maintenance logs.
- External data: IoT devices, social media, regulatory datasets.

2. Data Ingestion

- Use ETL (Extract, Transform, Load) pipelines or modern ELT frameworks.
- Streaming platforms (Apache Kafka, AWS Kinesis) for real-time ingestion.
- APIs for external or partner data.

3. Data Transformation & Cleaning

- Standardize formats (e.g., unify timestamps, normalize identifiers).
- Handle missing values and duplicates.
- Apply telecom-specific rules (e.g., mapping cell IDs to geographic locations).

4. Data Integration Layer

- Data warehouse (Snowflake, BigQuery, Redshift) for structured analytics.
- Data lake (Hadoop, Azure Data Lake, AWS S3) for raw/unstructured data.
- Lakehouse architectures (Databricks, Delta Lake) for flexibility.

5. Analytics & Insights

- Customer behavior analysis (usage patterns, churn prediction).
- Network optimization (capacity planning, fault detection).
- Fraud detection (SIM cloning, unusual call patterns).
- Revenue assurance (billing accuracy, leakage detection).

6. Governance & Compliance

- Ensure compliance with telecom regulations (e.g., TRAI in India, FCC in the US).
- Apply strict data privacy and security measures (GDPR, local laws).
- Metadata management and lineage tracking.

